

DEVIXO SOLUTIONS
AI/ML INTERNSHIP PROGRAM

TASK 01

Python Programming, Data Analysis & Visualization

Video Game Sales Analysis

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Domain	AI/ML
Task	Task 01
Dataset	Video Game Sales - vgsales.csv
Submission Date	13 September 2026

GitHub Repository: <https://github.com/zohaakhawajaa/devixo-solutions-ai-ml-internship/tree/main/Task-01>

1. Introduction

This project analyzes historical video game sales data to understand sales patterns across years, platforms, genres, publishers, and regions. The analysis was completed in Python using Pandas, NumPy, Matplotlib, and Seaborn.

2. Objective

The objective is to follow a complete data-analysis workflow: load and understand the dataset, identify data-quality issues, clean the data, perform exploratory and statistical analysis, create visualizations, and record findings.

3. Problem Statement

The dataset contains thousands of game records with sales reported for different regions. The purpose of the analysis is to compare these records and identify useful patterns in game releases and sales performance.

4. Technologies Used

Tool / Library	Purpose
Python 3	Programming language
Pandas	Loading, cleaning, grouping, and analysis
NumPy	Numerical operations
Matplotlib	Plot creation and figure handling
Seaborn	Statistical and categorical visualizations
Jupyter Notebook / Google Colab	Development and execution

5. Dataset

The selected dataset is the Video Game Sales dataset stored in **vgsales.csv**. It contains 16,598 rows and 11 columns. The dataset includes game information, release year, platform, genre, publisher, regional sales, and global sales.

Column	Description
Rank	Ranking value
Name	Game title
Platform	Gaming platform
Year	Release year
Genre	Game genre
Publisher	Publisher name
NA_Sales	North American sales (millions)
EU_Sales	European sales (millions)
JP_Sales	Japanese sales (millions)
Other_Sales	Other regional sales (millions)
Global_Sales	Worldwide sales (millions)

6. Data Loading and Understanding

```
1 df = pd.read_csv("vgsales.csv")
2 df
```

Code used to load the CSV file.

After loading the dataset, the notebook displays the first five rows, last five rows, a random sample, column names, data types, and dataset information. The dataset shape is **16,598 rows × 11 columns**.

```
1 import pandas as pd
2 import numpy as np
3 import matplotlib.pyplot as plt
4 import seaborn as sns
```

Libraries imported for the analysis.

7. Data Cleaning

The initial check found missing values in two columns. Year had 271 missing values (1.6327%) and Publisher had 58 missing values (0.3494%).

Column	Missing	Percentage
Year	271	1.6327%
Publisher	58	0.3494%

Publisher

Missing Publisher values were replaced with **Unknown**. This keeps the records without assigning an unsupported publisher.

```
1 ## Imputing missing values of publisher :
```

Notebook code related to Publisher handling.

Year

The missing Year values were not replaced with an assumed year. Instead, a separate DataFrame containing records with known years was used for time-based analysis. The Year column in that DataFrame was converted to integer.

```
1 df.columns = df.columns.str.lower()
2 df_year = df.dropna(subset=["year"]).copy()
3
4 df_year["year"] = df_year["year"].astype(int)
```

Preparing the time-analysis DataFrame.

Column names were converted to lowercase for consistency. Duplicate records were checked and none were found. All five sales columns were checked for negative values and none were found. The available years range from 1980 to 2020.

8. Descriptive Statistics

The descriptive statistics below were calculated for the five sales columns.

Measure	NA	EU	JP	Other	Global
Mean	0.2647	0.1467	0.0778	0.0481	0.5374
Median	0.08	0.02	0.00	0.01	0.17
Minimum	0.00	0.00	0.00	0.00	0.01
Maximum	41.49	29.02	10.22	10.57	82.74
Std. deviation	0.8167	0.5054	0.3093	0.1886	1.5550

Interpretation

Global_Sales has a mean of 0.5374 million and a median of 0.17 million. The difference between these values indicates a right-skewed distribution. The maximum global sales value is 82.74 million, while the minimum is 0.01 million.

The global-sales range is 82.73 million and the standard deviation is 1.5550 million. This reflects the large variation between ordinary games and a small number of very high-selling titles.

Correlation

Relationship	Correlation
NA vs Global	0.9410
EU vs Global	0.9028
NA vs EU	0.7677
Other vs Global	0.7483
JP vs Global	0.6118

North American sales have the strongest correlation with global sales in this dataset, followed by European sales.

9. Time-Based Analysis

Number of games released over time

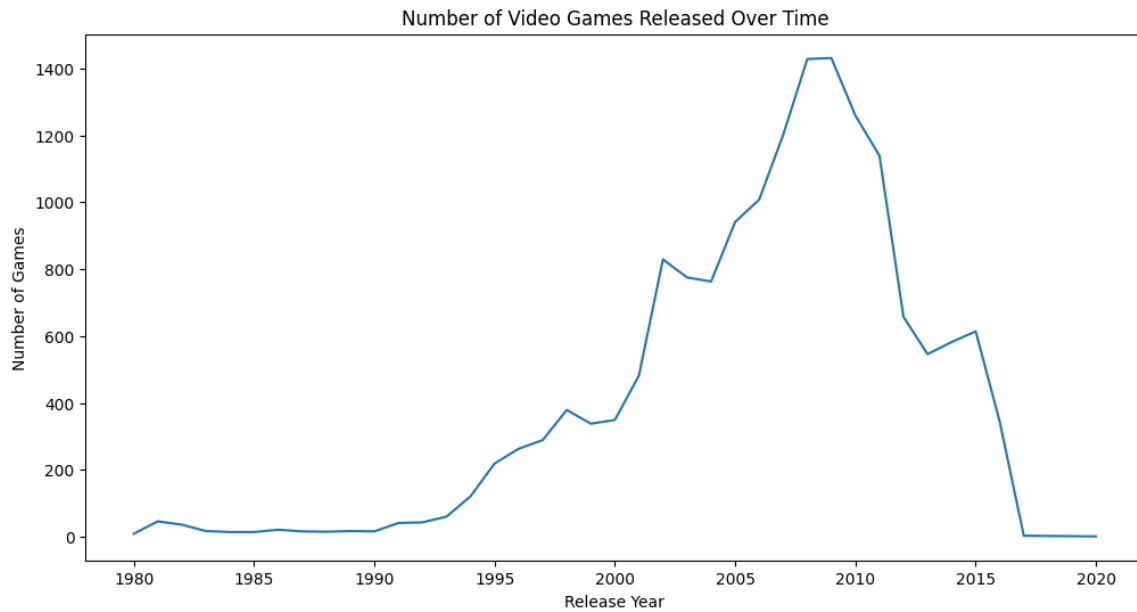


Figure 1. Number of video games released over time.

The number of releases increases considerably over the years and reaches its highest level around 2008–2009. The number then falls in later years.

Global sales over time

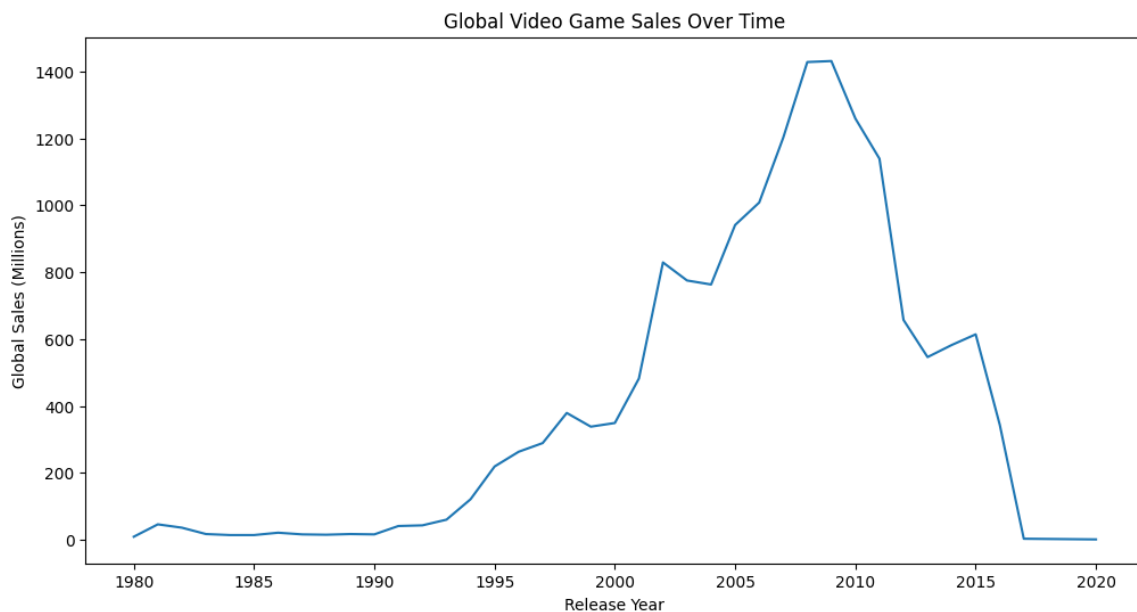


Figure 2. Total global sales by year.

Total global sales vary substantially between years. This measure is influenced by both the number of games released and their individual sales.

10. Platform and Genre Analysis

Top platforms by global sales

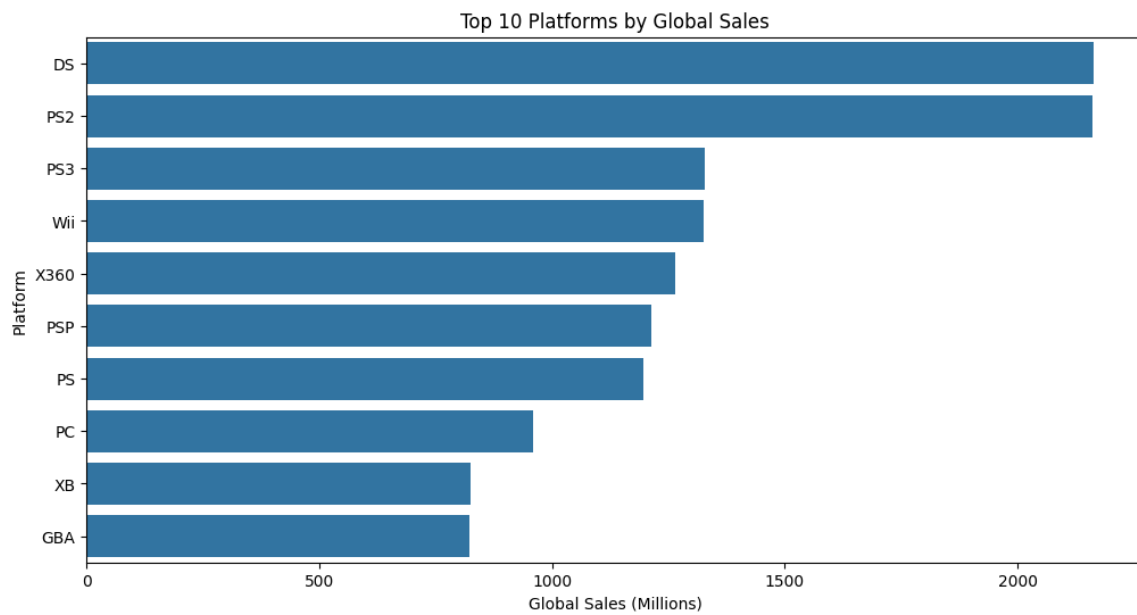


Figure 3. Top 10 platforms by global sales.

DS and PS2 are the two highest platforms in the displayed ranking by total global sales.

Global sales by genre

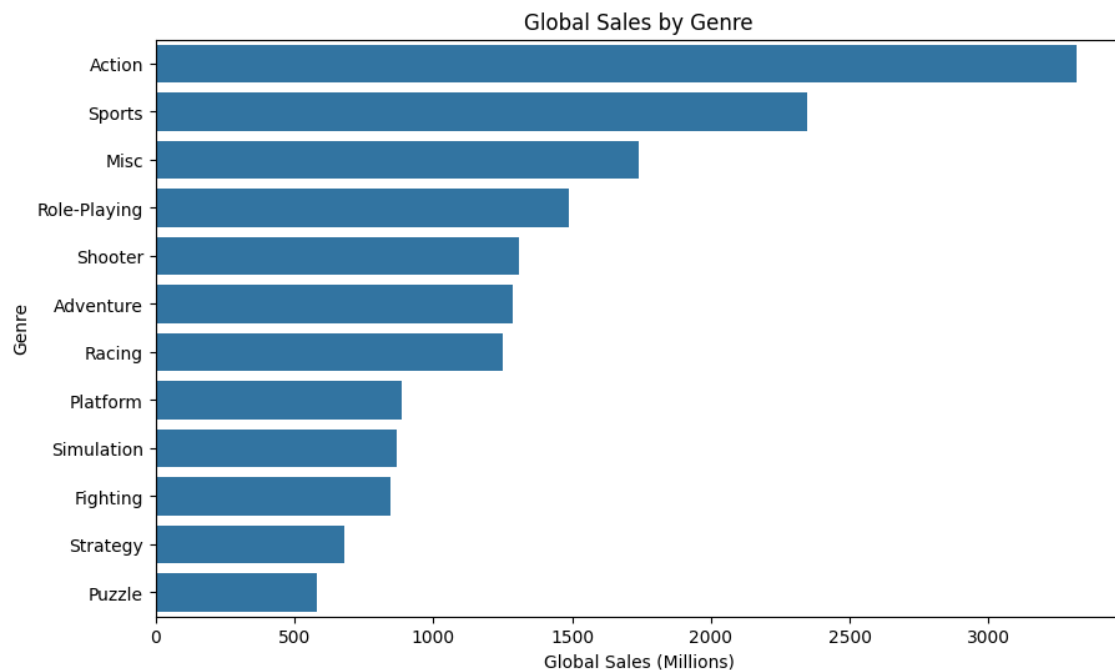


Figure 4. Global sales by genre.

Action has the highest total global sales among the genres, followed by Sports and Misc.

11. Distribution and Outlier Analysis

Genre distribution

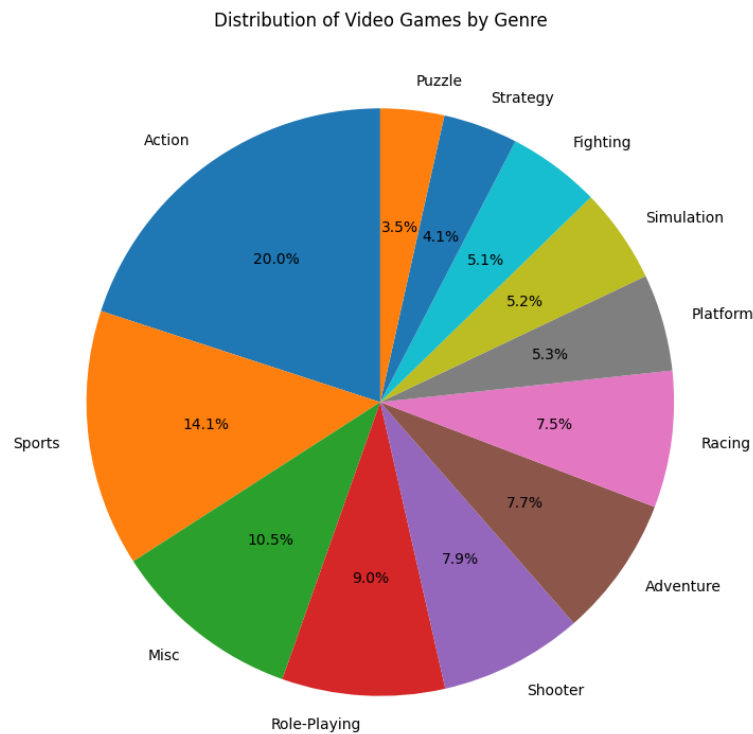


Figure 5. Distribution of games by genre.

The pie chart shows the proportion of records belonging to each genre. Action has the largest share of games in the dataset.

Histogram of global sales

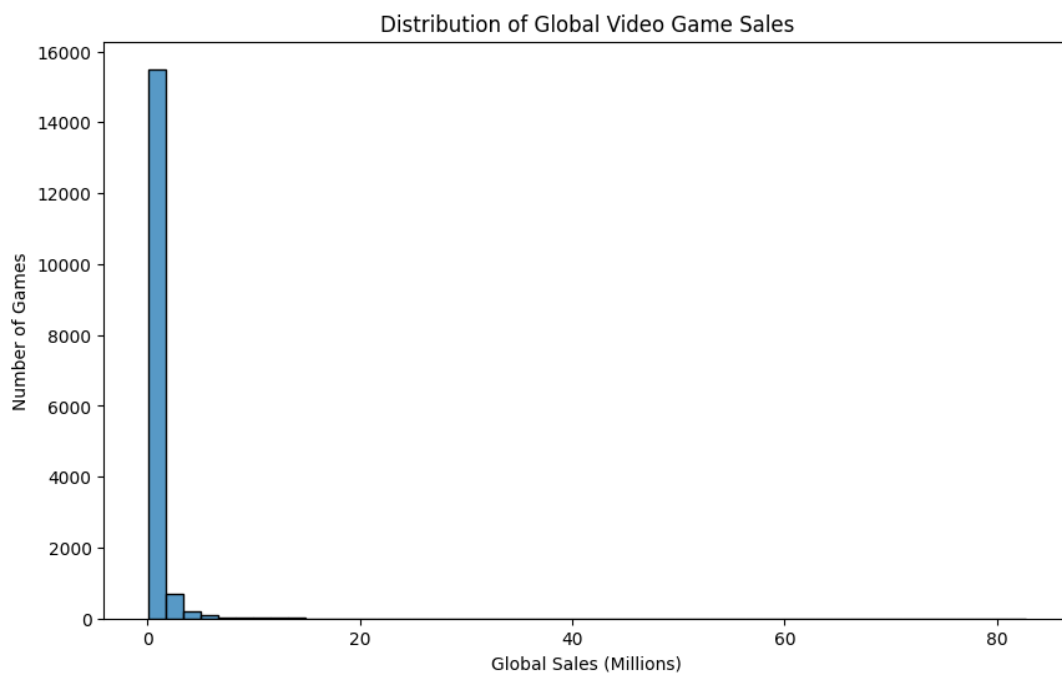


Figure 6. Distribution of global video game sales.

The histogram shows a strong right-skew. Most games have relatively low global sales, while a small number have much higher values.

Box plot

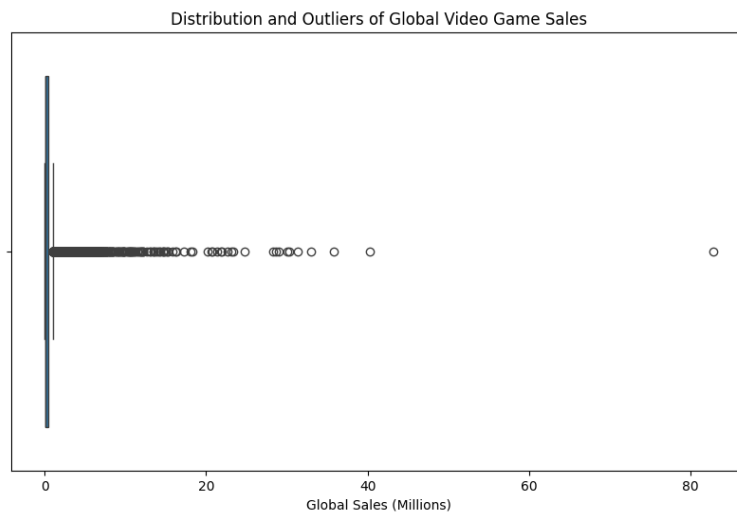


Figure 7. Global sales distribution and outliers.

12. Relationships Between Variables

North American vs European sales

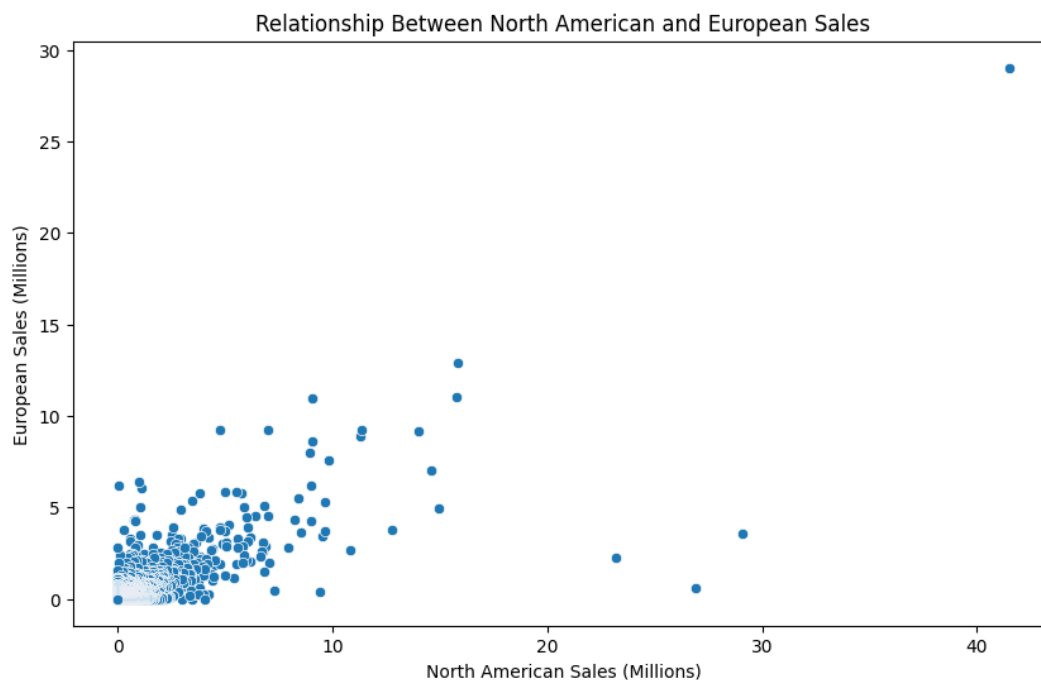


Figure 8. Relationship between North American and European sales.

The scatter plot shows a positive relationship between North American and European sales. Higher North American sales are generally associated with higher European sales.

Correlation heatmap

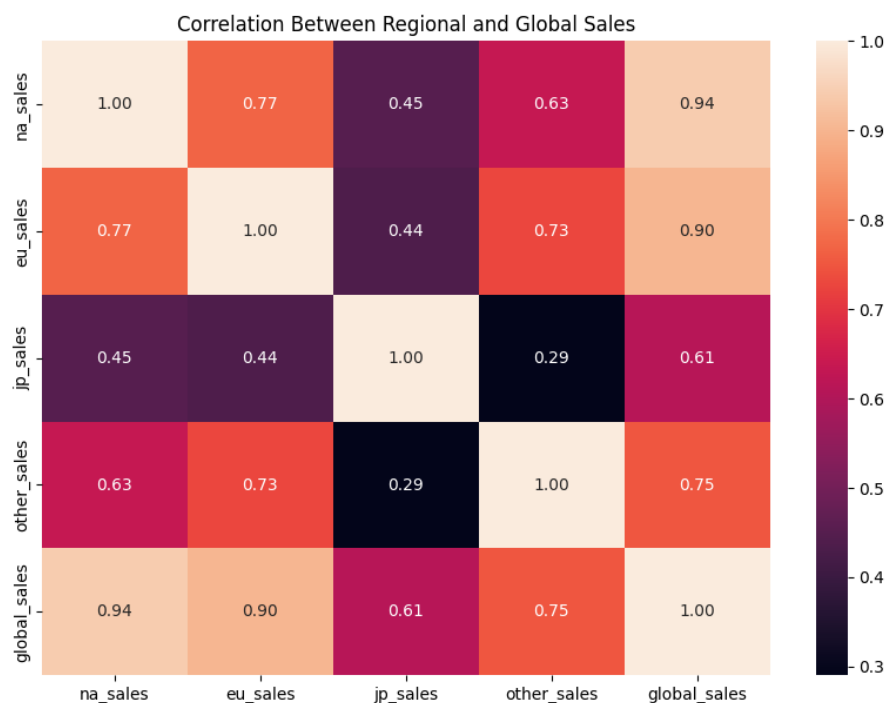


Figure 9. Correlation between regional and global sales.

The heatmap confirms the strongest relationship with global sales is for North American sales (0.9410), followed by European sales (0.9028). Japanese sales have a positive but weaker correlation (0.6118).

13. Additional Exploratory Analysis

Top platforms by number of games

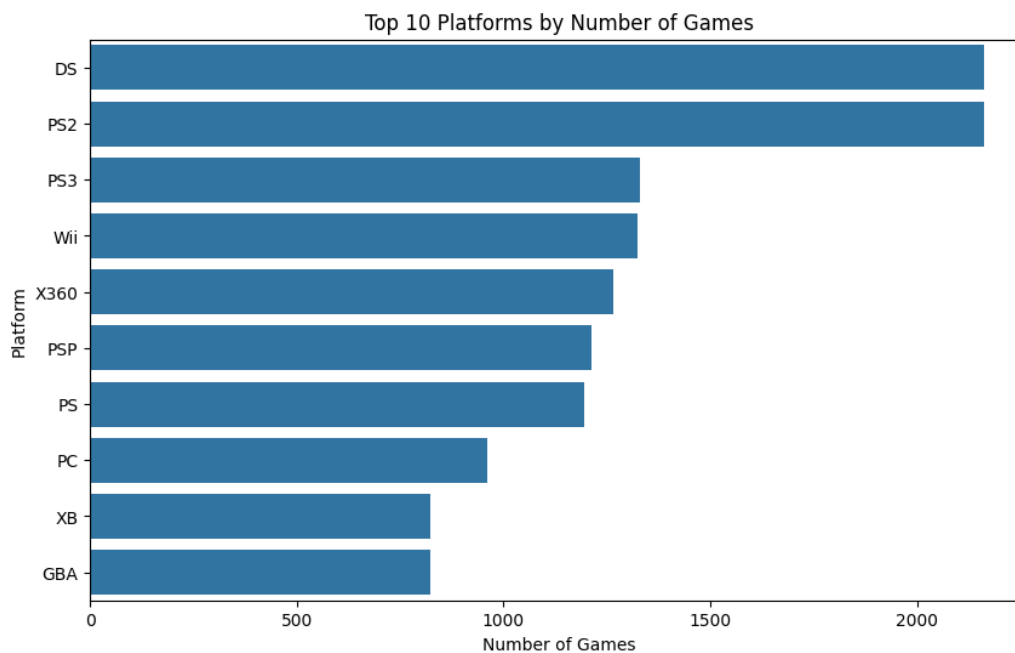


Figure 10. Top 10 platforms by number of games.

DS and PS2 contain the largest numbers of games among the platforms shown.

Sales distribution across common genres

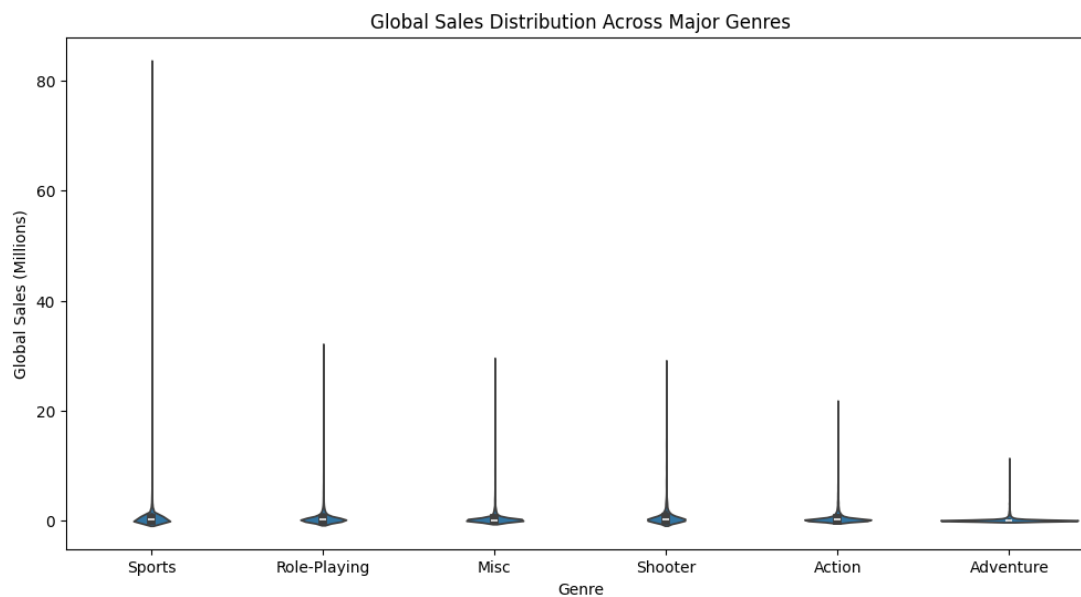


Figure 11. Violin plot for the six most common genres.

The distributions are concentrated near the lower end, with a small number of high-sales observations.

Pair plot

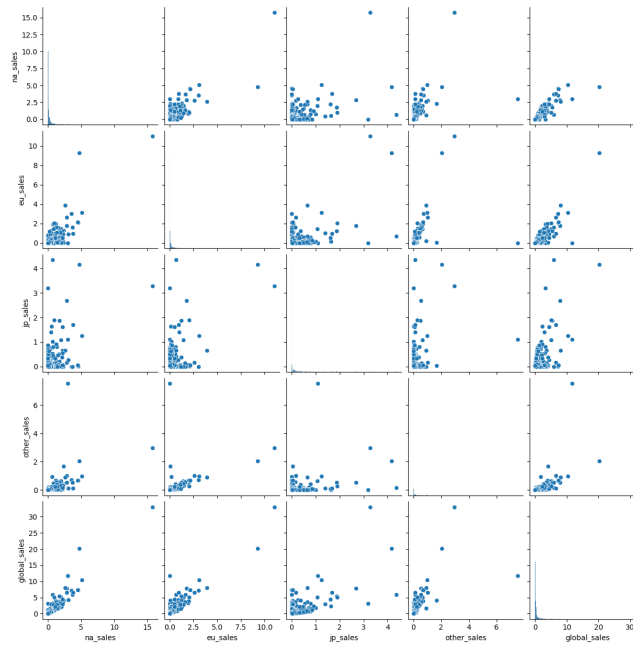


Figure 12. Pair plot of regional and global sales.

14. Publisher and Regional Analysis

Top publishers by global sales

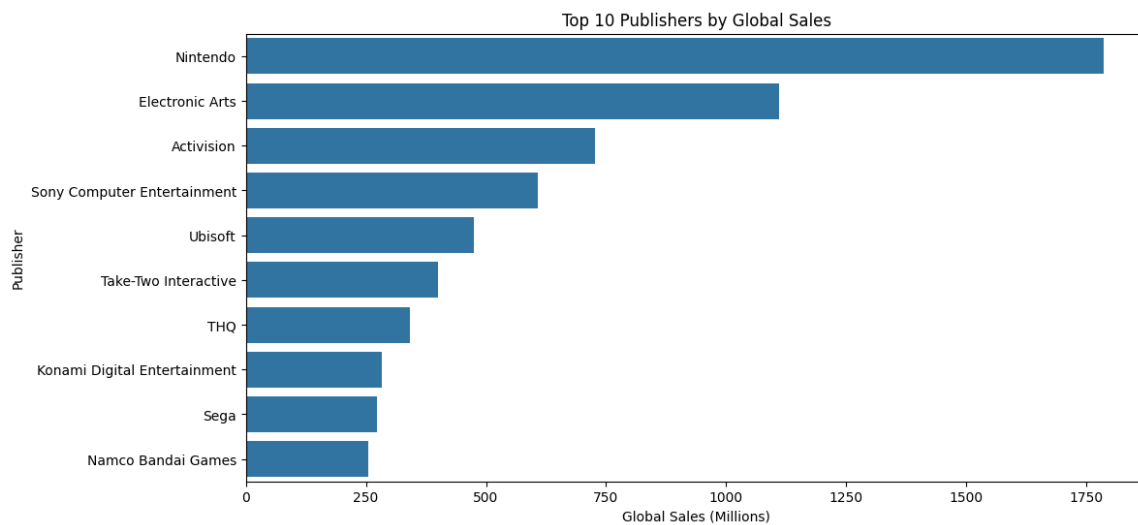


Figure 13. Top 10 publishers by global sales.

Publishers differ substantially in their total global sales. The result reflects both the size of their game catalogs and the performance of their games.

Total sales by region

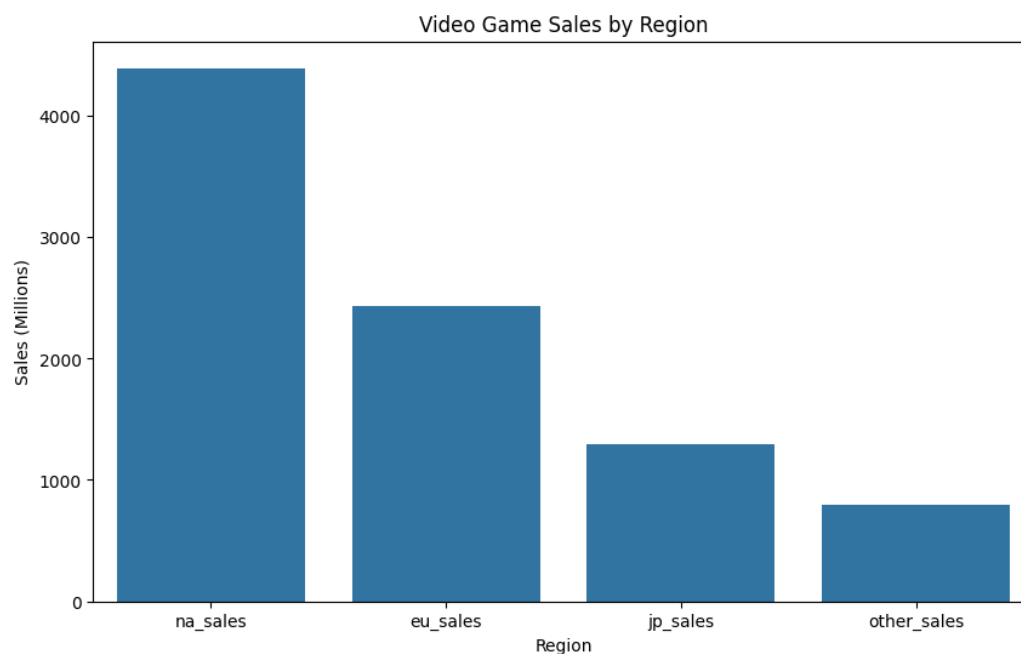


Figure 14. Total sales by region.

North America has the highest total regional sales in the dataset, followed by Europe, Japan, and other regions.

15. Top 10 Games

Rank	Game	Platform	Genre	Year	Global
1	Wii Sports	Wii	Sports	2006	82.74
2	Super Mario Bros.	NES	Platform	1985	40.24
3	Mario Kart Wii	Wii	Racing	2008	35.82
4	Wii Sports Resort	Wii	Sports	2009	33.00
5	Pokemon Red/Pokemon Blue	GB	Role-Playing	1996	31.37
6	Tetris	GB	Puzzle	1989	30.26
7	New Super Mario Bros.	DS	Platform	2006	30.01
8	Wii Play	Wii	Misc	2006	29.02
9	New Super Mario Bros. Wii	Wii	Platform	2009	28.62
10	Duck Hunt	NES	Shooter	1984	28.31

16. Major Insights

1. Wii Sports is the highest-selling game with 82.74 million global sales.
2. Nintendo appears repeatedly among the top-selling games.
3. The number of releases increases substantially over time and is highest around 2008–2009.
4. Global sales are right-skewed because most games have low sales and a small number have very high sales.
5. Median global sales (0.17 million) are much lower than mean global sales (0.5374 million).
6. North America has the highest total regional sales.
7. North American sales have the strongest correlation with global sales (0.9410).
8. European sales also have a strong correlation with global sales (0.9028).
9. Japanese sales have a positive but weaker correlation with global sales (0.6118).
10. DS and PS2 are the two most represented platforms by number of games.
11. Action has the highest total global sales among the genres.
12. No duplicate rows were found.
13. No negative values were found in the sales columns.

17. Challenges Faced

The main cleaning decision concerned the missing Year values. Filling these values with an assumed year could introduce information that is not present in the dataset, so the notebook used a separate DataFrame for time-based analysis and kept the original records.

The missing Publisher values were handled using the category **Unknown**. Another point requiring attention was the highly skewed sales distribution. The extreme values were kept because they are part of the actual dataset rather than automatically removing them as errors.

18. Conclusion

The analysis shows that video game sales are highly uneven. Most games have relatively low sales, while a small number of titles achieve very high global sales. There are also clear differences between platforms, genres, publishers, and regions.

North American and European sales have the strongest relationships with global sales in this dataset. Game releases increased considerably over time and reached their highest level around 2008–2009. The project provided practical experience with data loading, cleaning, EDA, statistical analysis, visualization, and interpretation using Python.

Submission Date: 13 September 2026